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The Choir Without a Conductor

Phase Transitions in Sparse Predictive Multi-Agent Synchronization

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github.com/hisenbug/kalman-kuramoto

1. Problem & Model

Can sparse communication synchronize adaptive agents? We replace all-to-all $\mathcal{O}(N^2)$ communication agents with **predictive-processing** ones that sample only a fraction ε of neighbours and blend the noisy result with their internal generative model via a Kalman update.

Kuramoto oscillators are the benchmark for collective synchronization in physics and biology, defined by N phase oscillators with distributed natural frequencies, coupled sinusoidally.

Kalman filter is an optimal recursive Bayesian estimator for linear-Gaussian systems that minimizes state covariance by integrating a transition-model prediction with noisy data.

Phase dynamics of each agent is driven by its own Kalman posterior over the mean field:

$$\frac{d\theta_i}{dt} = \omega_i + J\,\hat{r}_i \sin(\hat{\psi}_i - \theta_i), \qquad r\,e^{i\psi} = \frac{1}{N}\sum_{j=1}^{N} e^{i\theta_j} \tag{1}$$

Kalman posterior over frequency with prior variance $P = 1/K$, observation variance $R = \sigma^2/(\varepsilon N)$:

$$\hat{\omega}_{\text{post}} = \hat{\omega}_{\text{prev}} + K_{\text{gain}}\,(y - \hat{\omega}_{\text{prev}}), \qquad K_{\text{gain}} = \frac{1/K}{1/K + \sigma^2/(\varepsilon N)} \tag{2}$$

Two control variables: ε — bandwidth (fraction of neighbours sampled), K — precision (rigidity of priors).

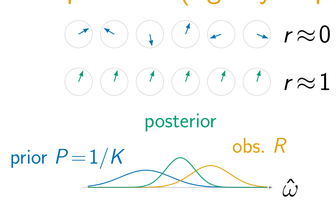

Figure 1: Phases θ_i disordered ($r \approx 0$) vs. aligned ($r \approx 1$); Kalman fuses prior and observation

2. Matched sync at ≈5% of baseline cost

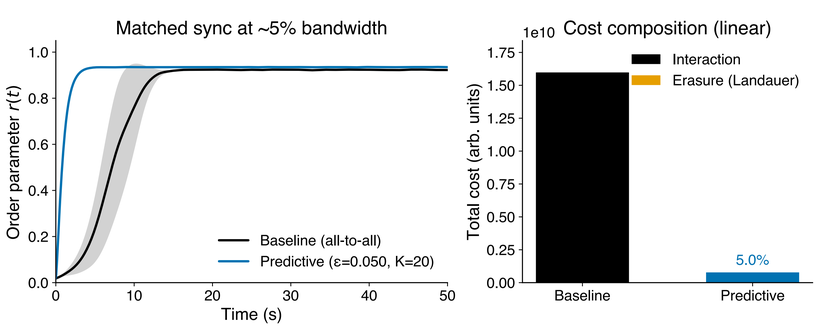

Figure 2: Left: order parameter $r(t)$; baseline (black) and predictive (blue) both reach $r_\infty \approx 0.93$. Right: total cost in arbitrarily scaled bits

Total predictive cost is 5% of baseline. **Communication saving is structural** to sparsity as erasure component is only ~3%.

Notation

N	number of agents	J	coupling strength
θ_i	phase of agent i	$r\,e^{i\psi}$	complex order parameter
ω_i	natural frequency, $\sim \mathcal{N}(0, \sigma_\omega^2)$	σ	observation noise
ε	sampling fraction per step	K	prior precision
$P = 1/K$	prior variance	$R = \sigma^2/(\varepsilon N)$	observation variance
y, $\hat{\omega}$	observation / estimate of frequency	r_∞	steady-state r
ε_c	critical sparsity	t_{90}	time to $0.9\,r_\infty$

3. Sharp phase transition at ε_c ; K-agnostic

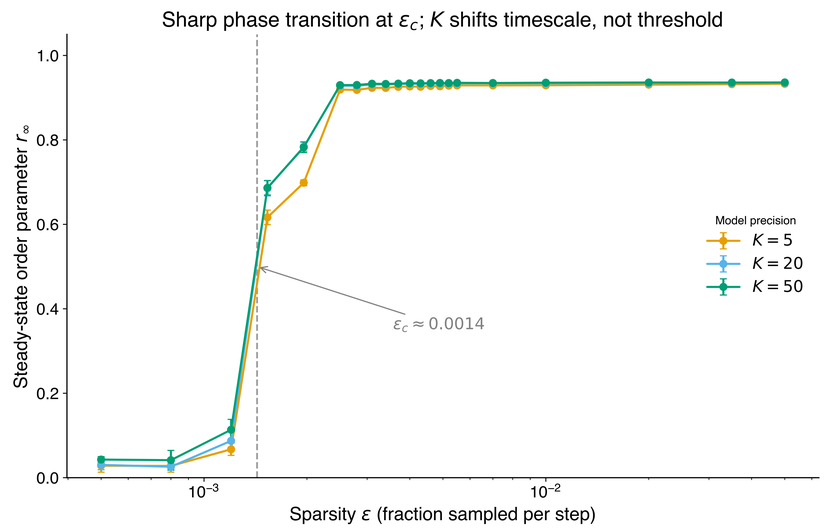

Figure 3: Steady-state r_∞ vs. bandwidth ε at three prior precisions $K \in \{5, 20, 50\}$. Vertical dashed line shows $\varepsilon_c \approx 0.0014$ ($N = 2000$)

Below ε_c, $r_\infty \approx 0.04$ regardless of K. **Synchronization is determined by sparsity (ε) alone;** K shifts only the time taken, not the threshold.

4. 2D landscape; K is only a timescale knob

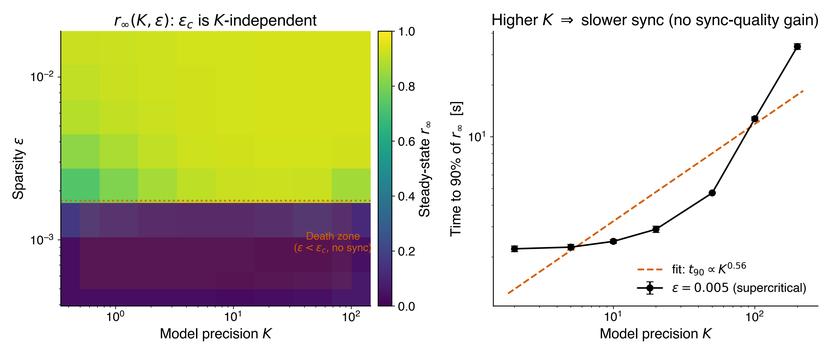

Figure 4: Left: heatmap of $r_\infty(K, \varepsilon)$ with a flat horizontal "death zone" at $\varepsilon < \varepsilon_c$ (red). Right: convergence time t_{90} vs. K at matched supercritical $\varepsilon = 0.005$.

The heatmap columns are *uniform in K*: precision does not change the phase-boundary. At matched r_∞, $t_{90} \propto K^{0.56}$ empirically; in the large-K asymptote this crosses over to linear Kalman settling ($t_{90} \propto K$), floored at small K by the Kuramoto phase timescale.

5. Finite-size scaling: sub-quadratic bandwidth

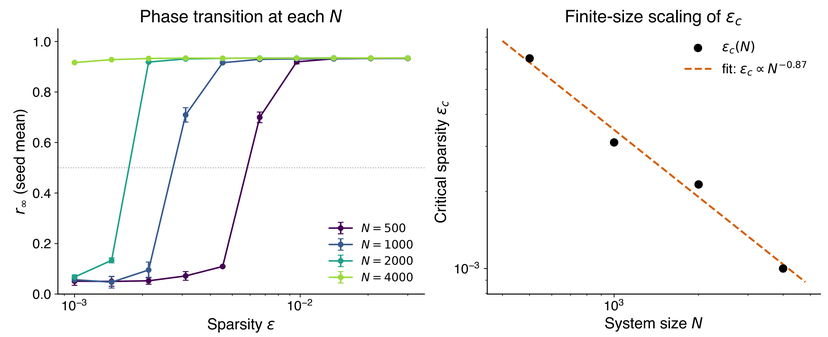

Figure 5: Left: $r_\infty(\varepsilon)$ for $N \in \{500, 1000, 2000, 4000\}$. Right: $\varepsilon_c(N)$ on log-log axes.

Empirical fit:

$$\varepsilon_c(N) \propto N^{-0.87} \quad \Longrightarrow \quad \varepsilon_c N^2 \propto N^{1.13}. \tag{3}$$

The fitted exponent is consistent, within 4-point scatter, with the mean-field prediction $\varepsilon_c N = \text{const}$. Either way, $\varepsilon_c N^2$ grows only slightly faster than linearly in N, hence the 5% ratio against the baseline is a fixed fraction, not a finite-N artifact.

6. Discussion & Future Work

A complex internal model is thermodynamically wasteful in a stationary environment. Erasure per step is K-flat, while sync time grows as $K^{0.56}$; total cost to sync therefore rises monotonically with K with no offsetting gain in steady-state r_∞.

In evolutionary r/K-strategy terms:

- Low K (wide prior, high Kalman gain) as **r-strategist**: reactive, cheap, and quick to track noisy observations.
- High K (narrow prior, low gain) as **K-strategist**: committed, slow, precise — metabolically justified only when the environment drifts on a timescale the prior can exploit.

Subsequent work will explore:

- Precision tests to separate -0.87 from the mean-field -1.
- Non-stationary $\omega_i(t)$: a drifting environment should expose a genuine Pareto frontier in K versus adaptation rate.
- Analytic derivation of the $t_{90}(K)$ crossover.

Predictive agents sync at ≈ 5% of the naive bandwidth.

References:

- *S. Strogatz*, "From Kuramoto to Crawford", Physica D 143, 1 (2000)
- *J. Acebrón et al.*, "The Kuramoto model: A simple paradigm for synchronization", Rev. Mod. Phys. 77, 137 (2005)
- *A. Anand*, "Coordination Requires Simplification: Thermodynamic Bounds on Multi-Objective Compromise in Natural and Artificial Intelligence", arXiv:2509.23144 (2025)

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